ISO 9001:2008 Certified Institute

JAVA INSTITUTE FOR ADVANCED TECHNOLOGY

Department of Examinations

Unit Name	Business Component Development I
Unit Id	JIAT/BCD I
Assignment Id	JIAT/BCD I/EX/01
Assignment Summary	Students will design, implement, and evaluate a Java EE–based enterprise application using components such as EJBs and messaging. The focus is on critical analysis, performance optimization, and meeting both functional and non-functional requirements. The assignment includes testing, performance validation, and documentation of design decisions and system improvements.
Duration	3 Weeks
Submission Via	Online (Student Portal)
Document Format	Document Format (Pdf)

GUIDELINES FOR CANDIDATES

- Your studies will be governed by the Java Institute Academic Regulations on Assessment, Progression and Awards.
- Students are expected to use reference books, the Internet, journals and other similar sources in order to accomplish the task specified above.
- Students are expected to refrain from repeating any content in their research document.
- At the re-assessment attempt, the mark is capped and the maximum mark that can be achieved is 40%.

CHEATING AND PLAGIARISM

Both cheating and plagiarism are totally unacceptable, and the Institute maintains a strict policy against them. It is YOUR responsibility to be aware of this policy and to act accordingly.

The basic principles are:

- Don't pass off anyone else's work as your own, including coding examples. This is plagiarism and is viewed extremely seriously by the Institute.
- Don't submit a piece of work, in whole or in part, that has already been submitted for assessment elsewhere. This is called duplication and, like plagiarism, is viewed extremely seriously by the Institute.
- Always acknowledge all the sources you have used in your assignment or project.
- If you are using the exact words of another person, always put them in quotation marks.

- Check whether the assignment is to be completed individually or if you are allowed to work with others.
- If you are doing group work, be clear about which parts you are expected to complete on your own.
- Never make up or falsify data to support your arguments.
- Never allow others to copy your work.
- Never lend disks, memory sticks, or copies of your coursework to any other student at the Institute, as this may lead to accusations of collusion

LEARNING OUTCOMES:

This assignment addresses the following unit learning outcomes:

1. Convey critical information and defining features of the Java Platform, Enterprise Edition (also known as Java EE) and demonstrate a clear understanding regarding the theories and terminology.

2. Identify and define the main concepts relating to implementing session beans as well as having a comprehensive understanding on evaluating what session bean is the most suitable to use at a given situation.

3. Undertake a critical analysis of the importance of JNDI in relation to the EJB components and its environment properties including how the dependency injection is used to access the session bean.

4. Clearly describe the bean lifecycle methods including the asynchronous communication methods that are introduced in this unit.

5. Critically describe singleton session bean life cycle and professionally apply the knowledge gained to develop Software Applications.

6. Demonstrate a critical understanding of the principles, terminology, and concepts pertaining to the Java Messaging System, evaluating the roles of the participant of the JMS API system.

7. Undertake a critical analysis of the properties, shortcomings and life cycle of the of message-driven beans while using object-oriented programming concepts to create and configure a message-driven bean.

Enterprise Java Development Assignment

E-Commerce Platform Modernization Project

<u>Assignment Overview:</u>
You are a senior software architect hired by "TechMart Online" to design a comprehensive modernization strategy for their e-commerce platform. The company needs to transition from their legacy system to a robust Java EE-based architecture that can handle enterprise-scale operations while implementing modern features like real-time notifications, advanced messaging, and scalable session management.

Business Scenario
TechMart Online is a growing e-commerce company facing critical scalability challenges:

Current State:

- Legacy monolithic system struggling with 1,000+ concurrent users
- Manual inventory management causing overselling issues
- Delayed order processing during peak sales periods
- No real-time customer notifications
- Poor integration with third-party services

Business Requirements:

- Support 10,000+ concurrent users with sub-second response times
- Real-time inventory synchronization across multiple warehouses
- Automated order processing with asynchronous notifications
- Comprehensive messaging system for internal and external communications
- Scalable session management for personalized shopping experiences

Technical Constraints:

- Must use Java EE platform and components
- Integration with existing database
- Cloud deployment capability
- 99.9% uptime requirement

Your Mission

Design a complete Java EE-based architecture that addresses all business requirements while demonstrating mastery of enterprise Java concepts. Your solution must include critical analysis of technology choices, comprehensive design decisions, and justification for architectural patterns with emphasis on Non-Functional Requirements (NFRs), component lifecycle optimization, and application performance strategies.

Comprehensive Design Requirements

Your architecture must address enterprise functionality through integrated Java EE solutions demonstrating critical thinking and analytical evaluation of:

- **Java EE Platform Critical Analysis:** Evaluate Java EE platform selection through comparative analysis with alternative enterprise frameworks, assess theoretical foundations supporting enterprise application development, critically examine platform limitations and propose mitigation strategies, and analyse how Java EE components address scalability and reliability requirements.

- **Session Bean Architecture Optimization:** Design stateless, stateful, and singleton session beans with critical evaluation of lifecycle implications on system performance, analyse which session bean type optimally addresses specific business functions, develop comprehensive lifecycle callback strategies with performance optimization, and critically assess resource utilization patterns across different bean types.

- **JNDI and Dependency Management Strategy:** Critically analyse JNDI importance in EJB component architecture for maintainability and scalability, evaluate dependency injection strategies with performance considerations, assess environment property management across deployment contexts, and analyse performance implications of service lookup versus injection patterns with optimization recommendations.

- **Asynchronous Communication Analysis:** Implement asynchronous methods using @Asynchronous annotations with critical evaluation of reliability implications, design Future object handling with timeout optimization strategies, analyse error handling and failure recovery approaches, compare synchronous versus asynchronous processing for different business scenarios, and evaluate performance trade-offs in concurrent processing.

- **Java Messaging System Architecture:** Design JMS-based messaging solution with critical evaluation of participant roles and responsibilities, implement point-to-point and publish/subscribe patterns with performance analysis, critically assess message delivery guarantees and business implications, analyse scalability limitations and propose optimization strategies, and evaluate monitoring approaches for messaging system reliability.

- **Message-Driven Bean Implementation:** Create message-driven bean solutions with critical analysis of component lifecycle efficiency, evaluate MDB properties impact on system performance, identify limitations in high-throughput contexts with mitigation strategies, analyse lifecycle management optimization across deployment scenarios, and apply object-oriented principles with performance considerations.

Non-Functional Requirements Analysis

Your solution must demonstrate critical thinking in addressing key NFRs:

- **Performance Analysis:** Evaluate response time optimization strategies, analyse throughput enhancement approaches, assess resource utilization patterns, and propose performance monitoring mechanisms.

- **Scalability Evaluation:** Analyse scaling strategies for Java EE components, evaluate load balancing considerations, assess database scaling approaches, and propose auto-scaling mechanisms.

- **Reliability Assessment:** Evaluate fault tolerance strategies, analyse disaster recovery approaches, assess circuit breaker patterns for external services, and propose comprehensive availability measures.

Project Development Requirements

Develop a working prototype demonstrating practical application of Java EE technologies with measurable performance characteristics.

Core Implementation Requirements:

- Multiple session bean types with business logic and lifecycle optimization
- JNDI configuration and dependency injection with performance monitoring
- Asynchronous methods with error handling and performance measurement
- JMS messaging system with throughput optimization
- Message-driven beans with lifecycle optimization
- Database integration with connection optimization
- Web interface with performance metrics display

Technology Stack:

- Java EE 8+ (Jakarta EE acceptable)
- Application Server: WildFly, GlassFish/Payara, or TomEE
- Database: PostgreSQL/MySQL with connection pooling
- Build Tool: Maven
- Testing: JUnit 5 + Arquillian + performance testing tools

Deliverables

1. **Working Project Implementation**

Source Code Package:

- Complete project with performance monitoring implementation
- Database schema with optimization configurations
- Configuration files with performance-focused settings
- Deployment guide with optimization instructions
- Performance measurement and monitoring setup

2. **Technical Implementation Documentation (1,500-2,000 words)**

Architecture Documentation:

- System architecture with component relationship analysis
- Implementation specifications with optimization techniques
- Configuration documentation with performance tuning parameters
- Deployment architecture with performance considerations

Component Analysis:

- Session bean implementations with lifecycle management strategies
- Messaging system design with throughput optimization
- Database integration with connection optimization
- Asynchronous processing with performance measurement results

Implementation Analysis with Critical Evaluation:

- Critical analysis of Java EE platform selection with comparative evaluation
- Session bean design decisions with lifecycle optimization analysis
- JNDI and dependency injection strategy with performance impact assessment
- Asynchronous processing implementation with reliability analysis
- JMS messaging architecture with scalability evaluation
- Message-driven bean design with lifecycle optimization analysis

3. Critical Analysis and Test Report (1,000-1,500 words)

Testing Strategy and Performance Validation:

- Unit testing approach with performance test cases
- Integration testing with load scenarios
- Performance benchmarking with optimization results
- Critical evaluation of testing results with improvement recommendations
- NFR validation through systematic testing approach

Critical Reflection and Optimization:

- Analysis of design alternatives with justification for chosen approaches
- Performance bottleneck identification with resolution strategies
- Scalability analysis with specific optimization recommendations
- Critical assessment of implementation limitations with mitigation strategies

-END OF ASSIGNMENT-

Evaluation Criteria

- **Critical Analysis and Technical Understanding (40 marks)**
  - **In-depth Evaluation of Java EE Platform (10 marks)**
    Demonstrates critical thinking in evaluating strengths, limitations, and suitability of Java EE for enterprise development.
  - **Comparative Analysis and Design Justification (10 marks)**
    Logical comparison between alternative technologies/frameworks and well-reasoned justification of chosen architecture/design.
  - **Lifecycle Management and Component Optimization (10 marks)**
    Clear understanding and application of Java EE component lifecycles with appropriate optimizations.
  - **Non-Functional Requirements (NFR) Analysis and Mitigation Strategies (10 marks)**
    Identifies relevant NFRs (e.g., scalability, availability) and provides effective mitigation strategies.
- **Working Implementation Quality (35 marks)**
  - **Functional Java EE Application with Performance Monitoring (10 marks)**
    Complete implementation of functional application with integrated performance monitoring tools/techniques.
  - **Correct Implementation of Required Components (8 marks)**
    Proper use and integration of EJBs, messaging, etc., as per assignment requirements.
  - **Code Quality and Optimization (9 marks)**
    Clean, maintainable, and well-documented code with implemented performance and resource optimizations.
  - **Deployment and Runtime Performance Measurement (8 marks)**
    Successful deployment with runtime benchmarks, profiling, or metrics collection demonstrating application performance.
- **Testing and Validation (25 marks)**
  - **Comprehensive Testing Strategy with Performance Emphasis (8 marks)**
    Clearly defined testing plan including unit, integration, and performance test cases.
  - **Test Execution and Results Analysis (6 marks)**
    Evidence of tests being executed with meaningful analysis of results.
  - **Performance Optimization Evaluation (6 marks)**
    Critical reflection on system performance, bottlenecks, and implemented enhancements.
  - **Testing Documentation and Recommendations (5 marks)**
    Well-structured documentation including tools used, methodologies applied, and future improvement suggestions.

Assignment Submission Guidelines for Students

- **Format:**
    - Assignment Documents Rules:
        - Paper Size: A4
        - Line Spacing:1.5
        - Header and Footer:1 inch
        - Basic Font Size:11pt
        - Heading:14pt
        - Sub-headings:12pt, Bold
        - Body:11pt, Justified Aligned
        - Font Style: Times New Roman/Calibri
    - Provide the following information in their assignment coversheet:
        - Student name
        - NIC No
        - Subject name
        - Subject code
        - Branch
    - Working project as compressed archive with complete implementation
    - Combined documentation as single PDF (2,500-3,500 words total)
    - Performance monitoring results and optimization evidence
- **Code Requirements:** Must compile, deploy, and demonstrate performance optimization
- **Individual Work:** Original implementation and critical analysis required
- **Resources:** Official Java EE documentation and academic sources with proper citation
- **Referencing:** Must be cited using the Harvard referencing system, with a reference list included at the end.

Submission Checklist:

- ☐ Complete source code with performance monitoring
- ☐ Deployable application with optimization configurations
- ☐ Database setup with performance tuning
- ☐ Comprehensive test suite with performance validation
- ☐ Critical analysis documentation with NFR evaluation
- ☐ Performance monitoring results and optimization evidence

Academic Integrity and Assessment Notes

Critical Thinking Emphasis: This assignment prioritizes critical analysis and evaluation over comprehensive implementation. Deep understanding of Java EE concepts with effective optimization strategies may receive higher marks than extensive implementations without critical analysis.

Performance and Optimization Focus: The systematic analysis of performance characteristics and optimization strategies demonstrates understanding of enterprise application development challenges. Quality of critical thinking and problem-solving approach is essential for success.

Real-World Application: Your solution should demonstrate practical understanding of enterprise development challenges with critical evaluation of design decisions, performance trade-offs, and optimization strategies suitable for production deployment scenarios.